Lab 8:Asynchronous and Synchronous Counters

CENG 2112.01 Lab for Digital Circuits

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Contents

Abstract

An asynchronous counter is a sequential digital circuit where every flip-flop after the first involved are not clocked simultaneously, rather the first flip-flop is driven by a clock pulse and each flip-flop in the circuit is triggered by the output of the flip-flop before it. A synchronous counter however, has all flip-flops sharing a clock signal. This allows all the bits in the flip-flops to change at the same time. These counters can be used in any high speed applications as well as precise timing in digital devices.

Introduction

A D-flip-flop (part # 7474) can be used in either an asynchronous or synchronous circuit, the amount of D-flip-flops needed is based on how many bits the circuit requires to make it work. One D-flip-flop is used for one bit, so if used in a digital clock with a seven-segment display then there will be seven D-flip-flops used. Since they can be used in both an asynchronous and synchronous circuit, a circuit can be built to make a counter up to any number up to ten.

Requirements

The requirement for this lab is to simulate an asynchronous and synchronous circuit to make a counter that counts up to five asynchronously and seven synchronously then returns to zero on Multisim, then we wired the circuits on a bread board to match what Multisim's output is.

Implementation

The first step in implementation was to build a circuit in Multisim to simulate an effective asynchronous and synchronous counter using D-flip-flops. The second step was to wire the circuits on a bread board using the LED lights on it to simulate a counter.

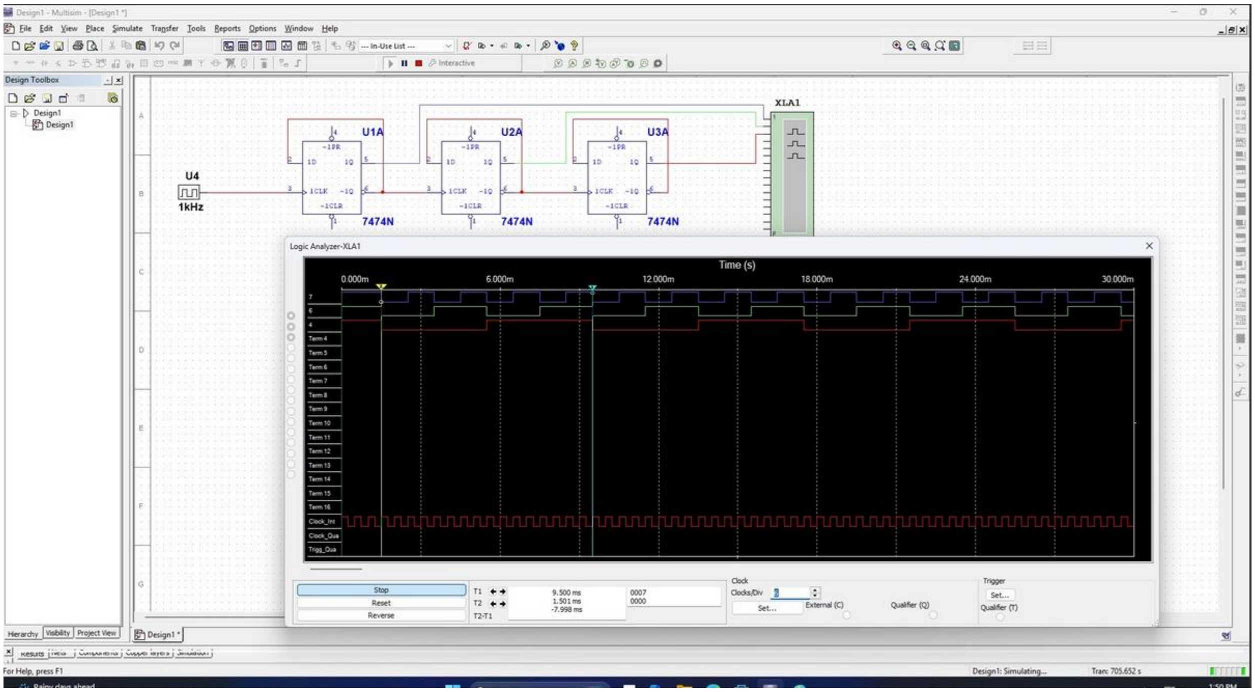

Figure 1.1: Asynchronous counter with logic analyzer to show it is effectively counting to 7 and resets to 0.

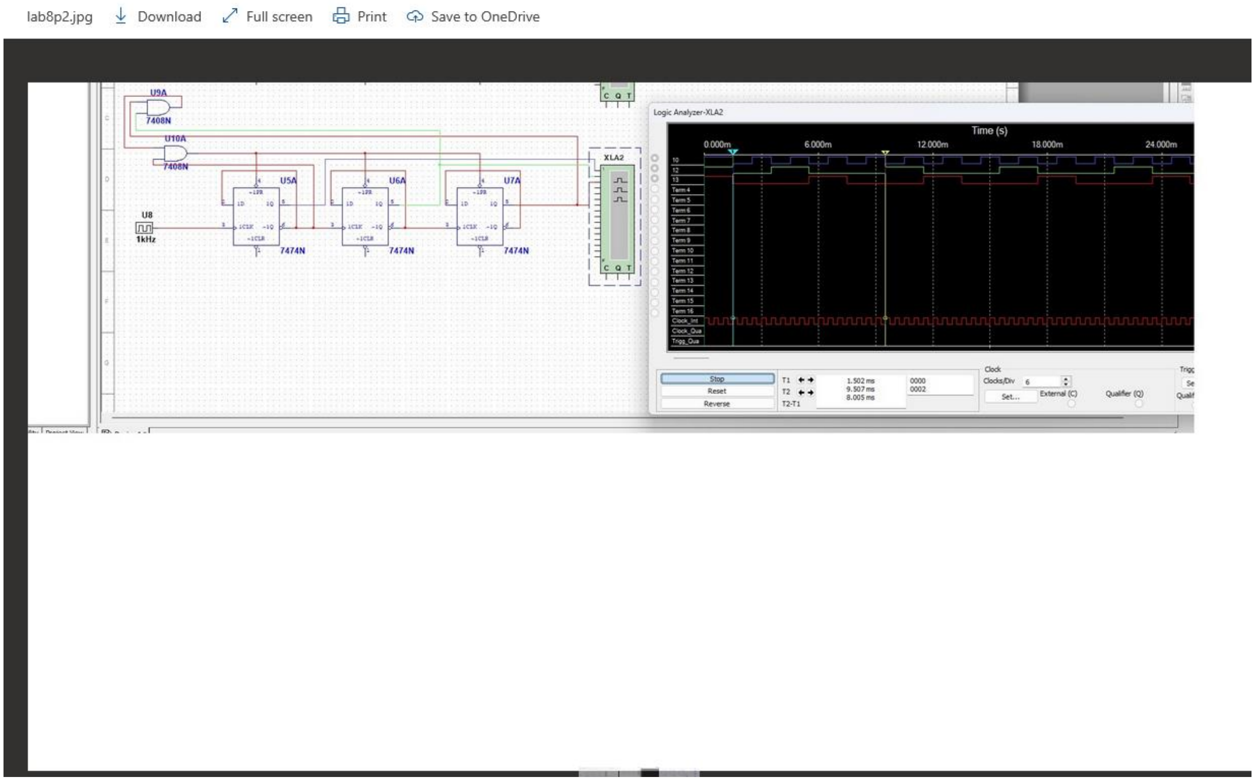

Figure 1.2: Synchronous counter with logic analyzer to show it is effectively counting to 7 and resets to 0.

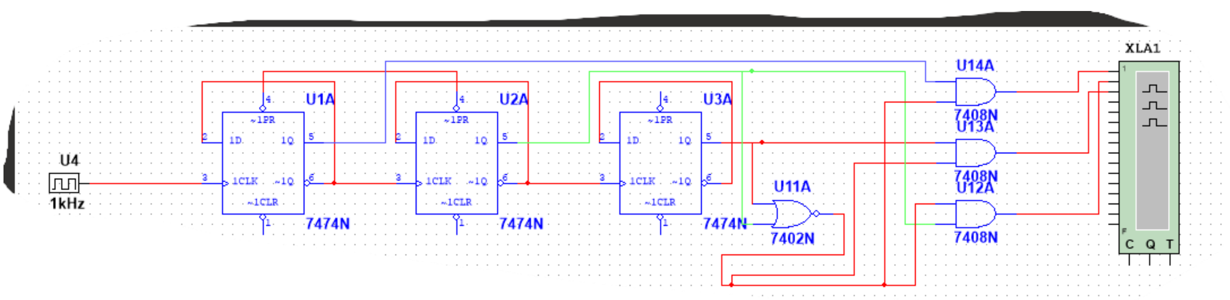

Figure 1.3: Asynchronous counter effectively counting to 5 and resets to 0.

We chose to go with these circuits because they effectively count to the desired numbers in an asynchronous/synchronous circuit. The reason why we went with three D-flip-flops is because we are ultimately dealing with three bits since the maximum number to count up to is seven.

The result in figure 2.1 shows the bread board using the asynchronous circuit with the leds showing a 0.

The result in figure 2.2 shows the bread board using the asynchronous circuit with the leds showing a 1.

The result in figure 2.3 shows the bread board using the asynchronous circuit with the leds showing a 2.

The result in figure 2.4 shows the bread board using the asynchronous circuit with the leds

showing a 3.
The result in figure 2.5 shows the bread board using the asynchronous circuit with the leds showing a 4.
The result in figure 2.6 shows the bread board using the asynchronous circuit with the leds showing a 5.
The result in figure 2.7 shows the bread board using the asynchronous circuit with the leds showing a 6.
The result in figure 2.8 shows the bread board using the asynchronous circuit with the leds showing a 7.

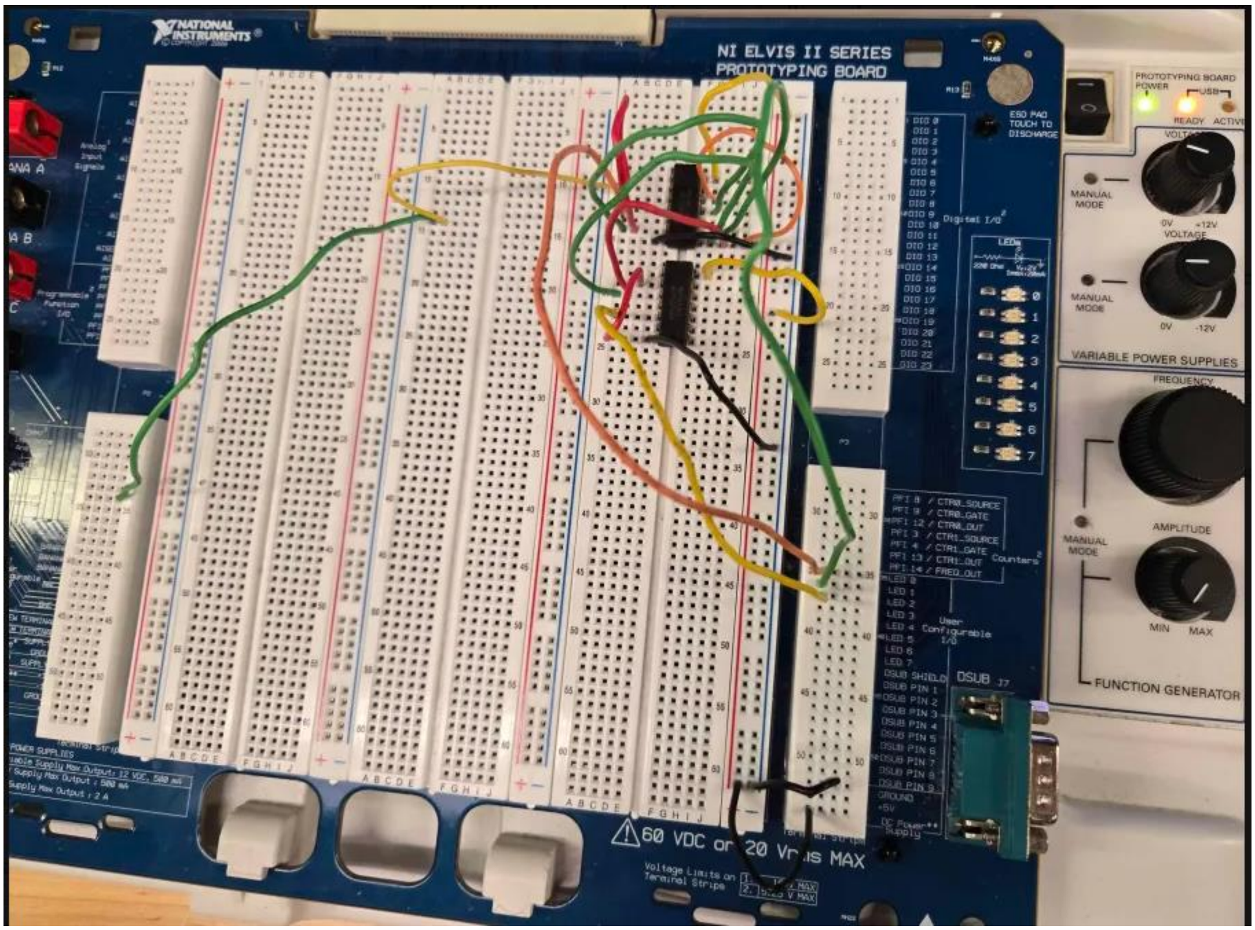

figure 2.1: asynchronous circuit showing a 0.

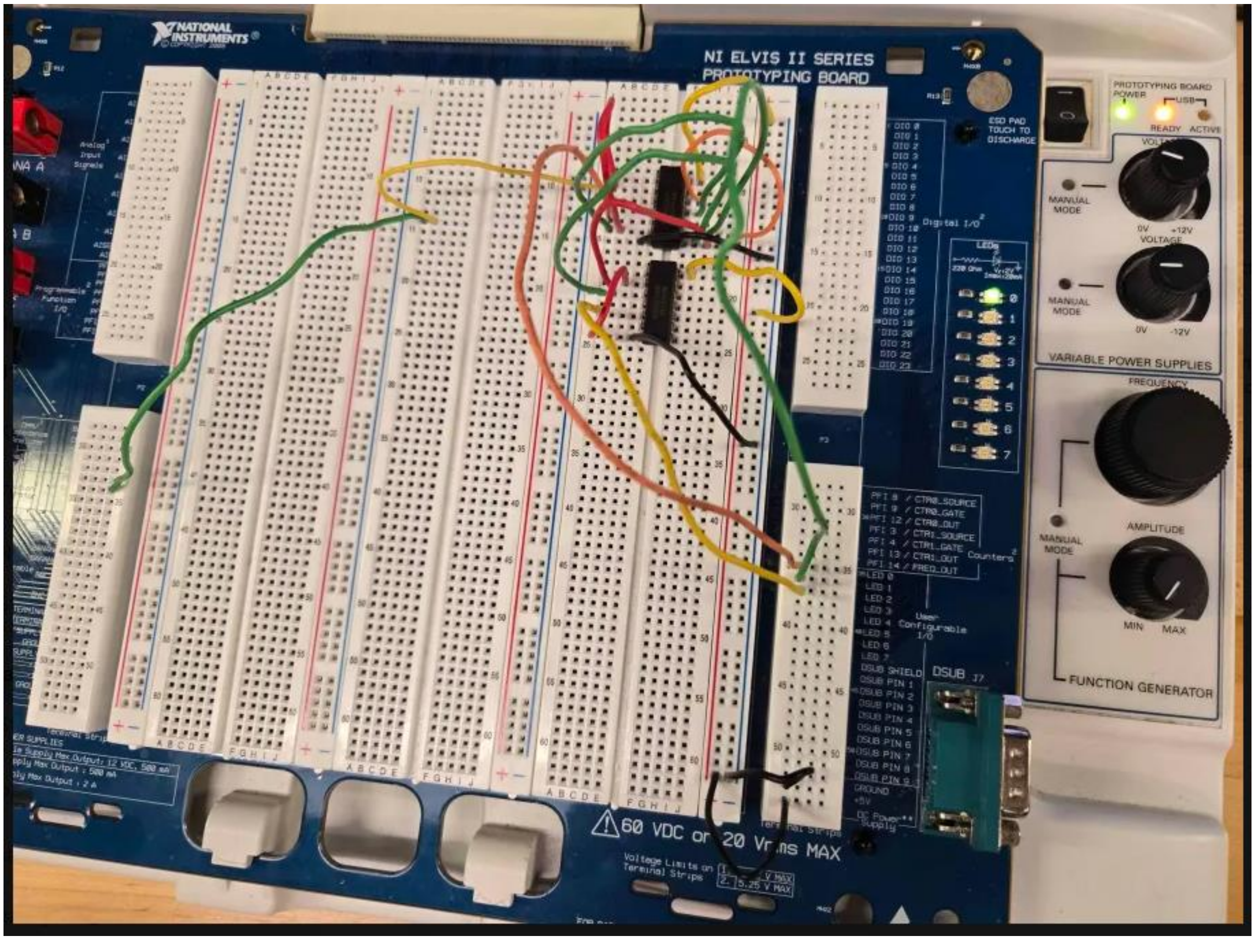

figure 2.2: asynchronous circuit showing a 1.

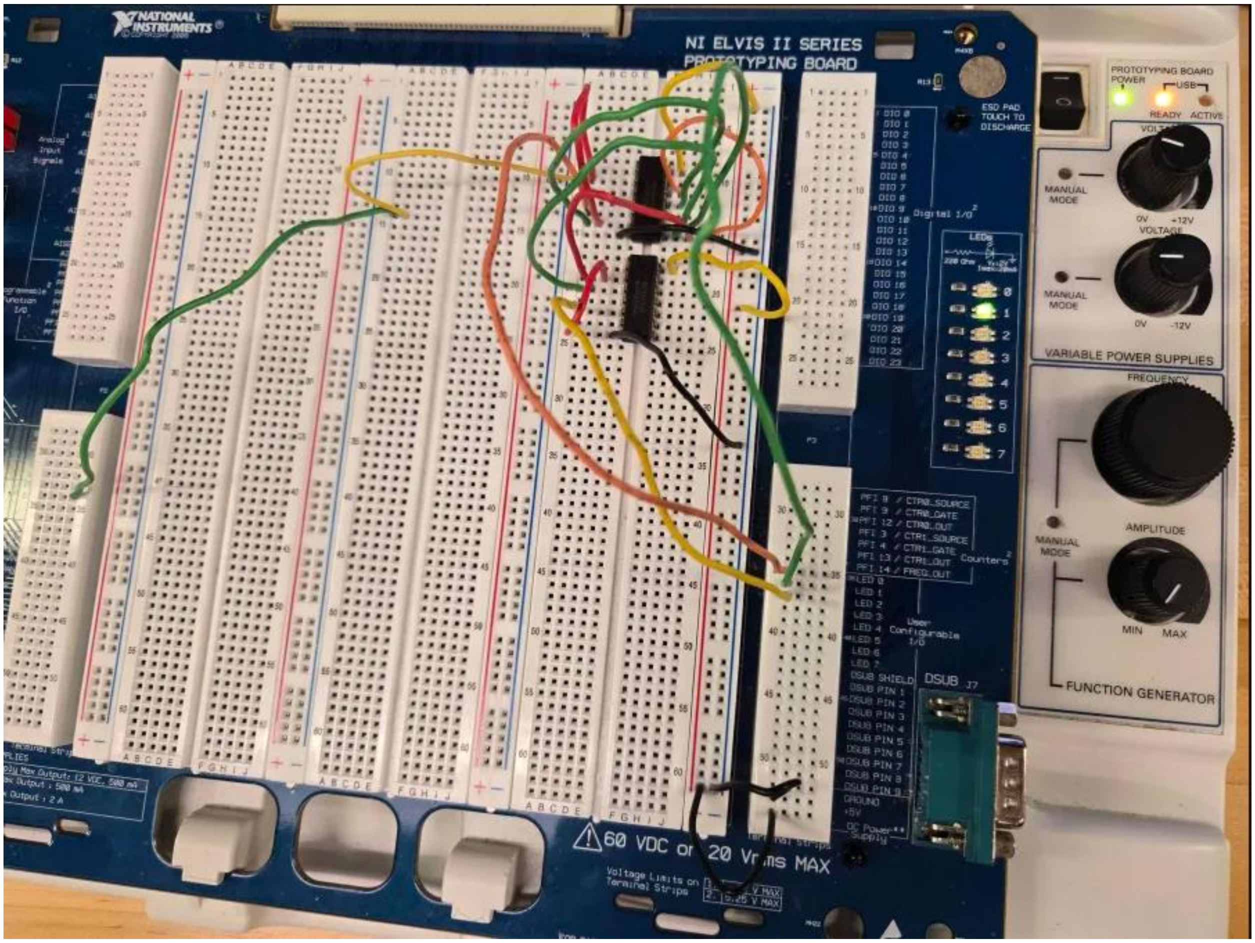

figure 2.3: asynchronous circuit showing a 2.

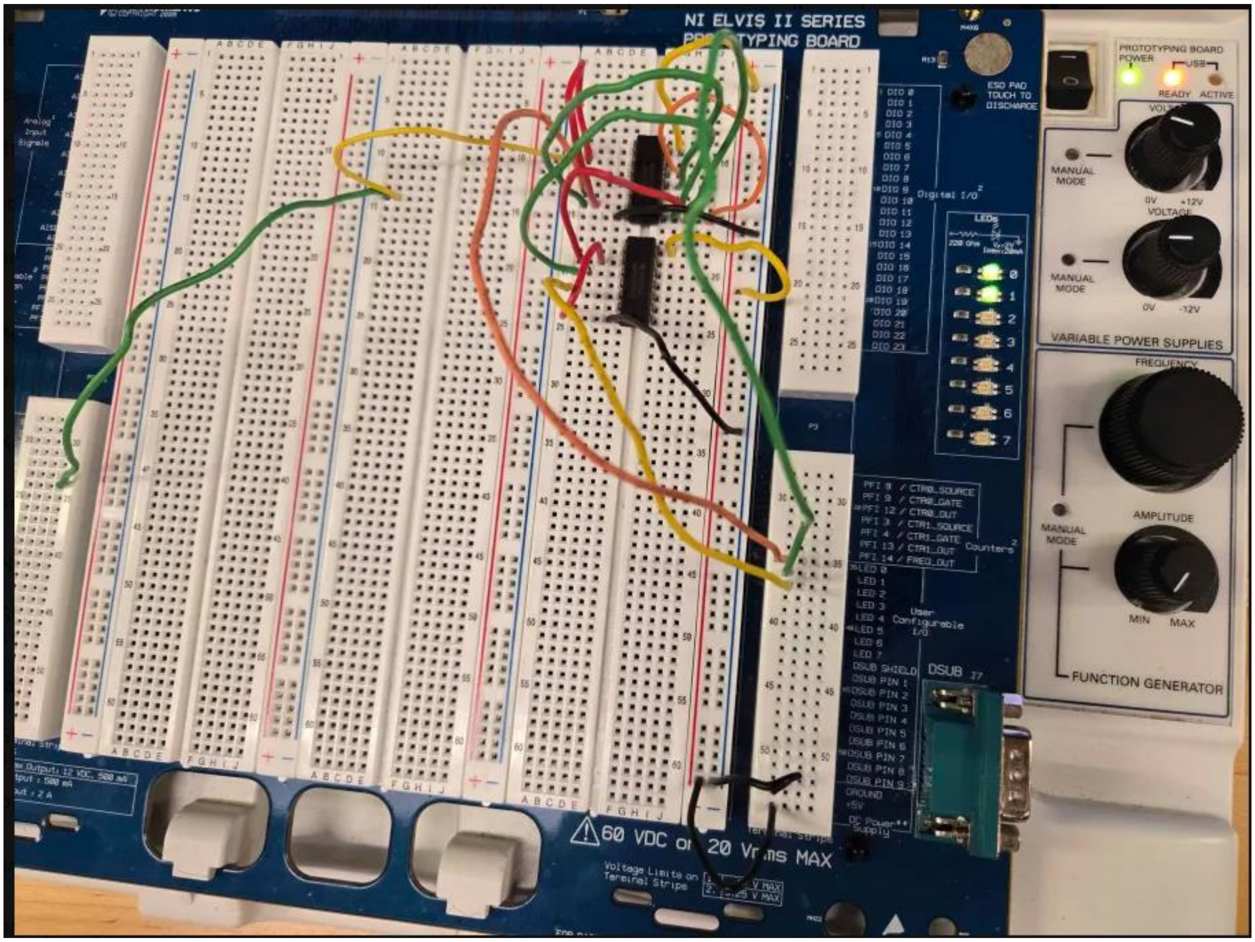

figure 2.4: asynchronous circuit showing a 3.

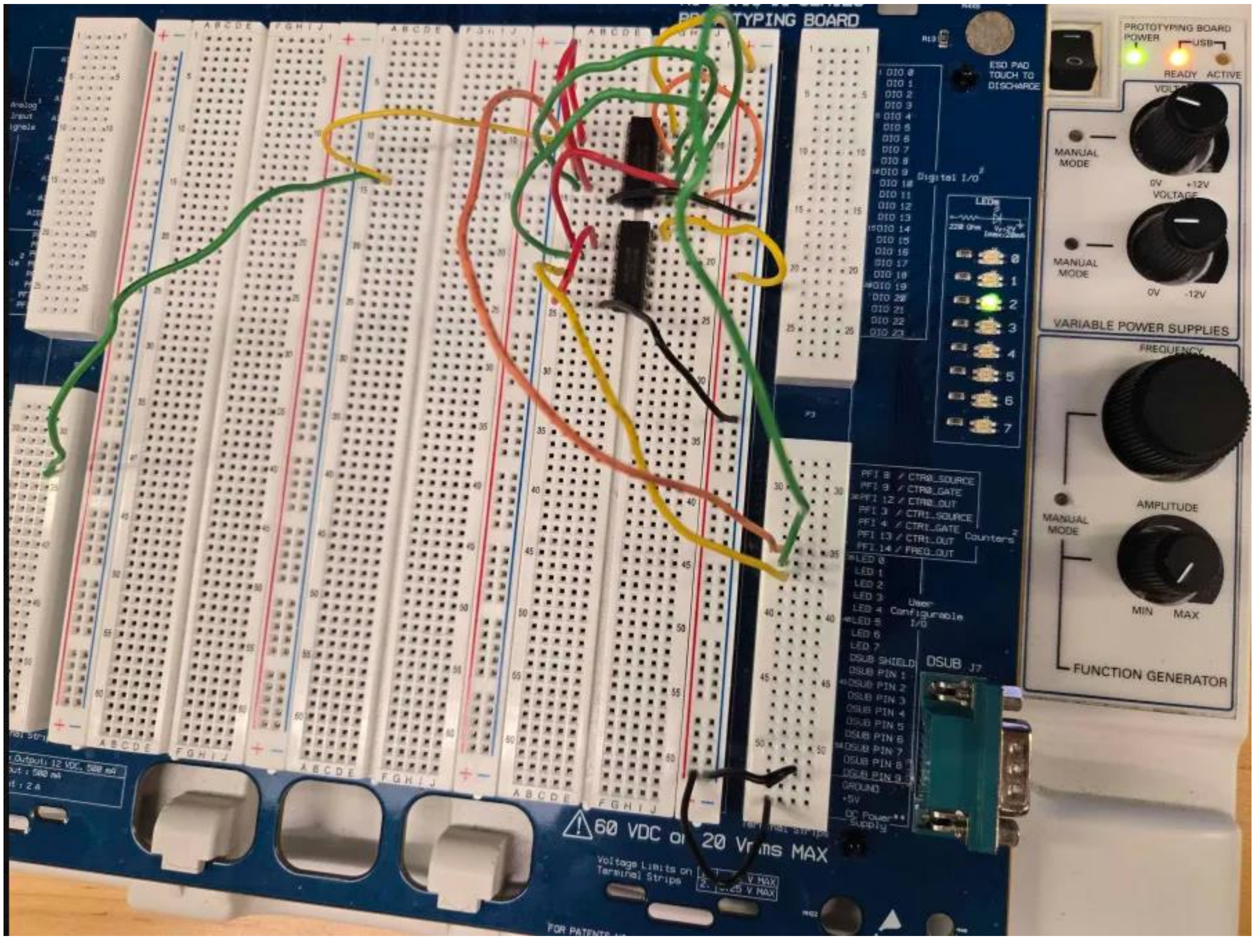

figure 2.5: asynchronous circuit showing a 4.

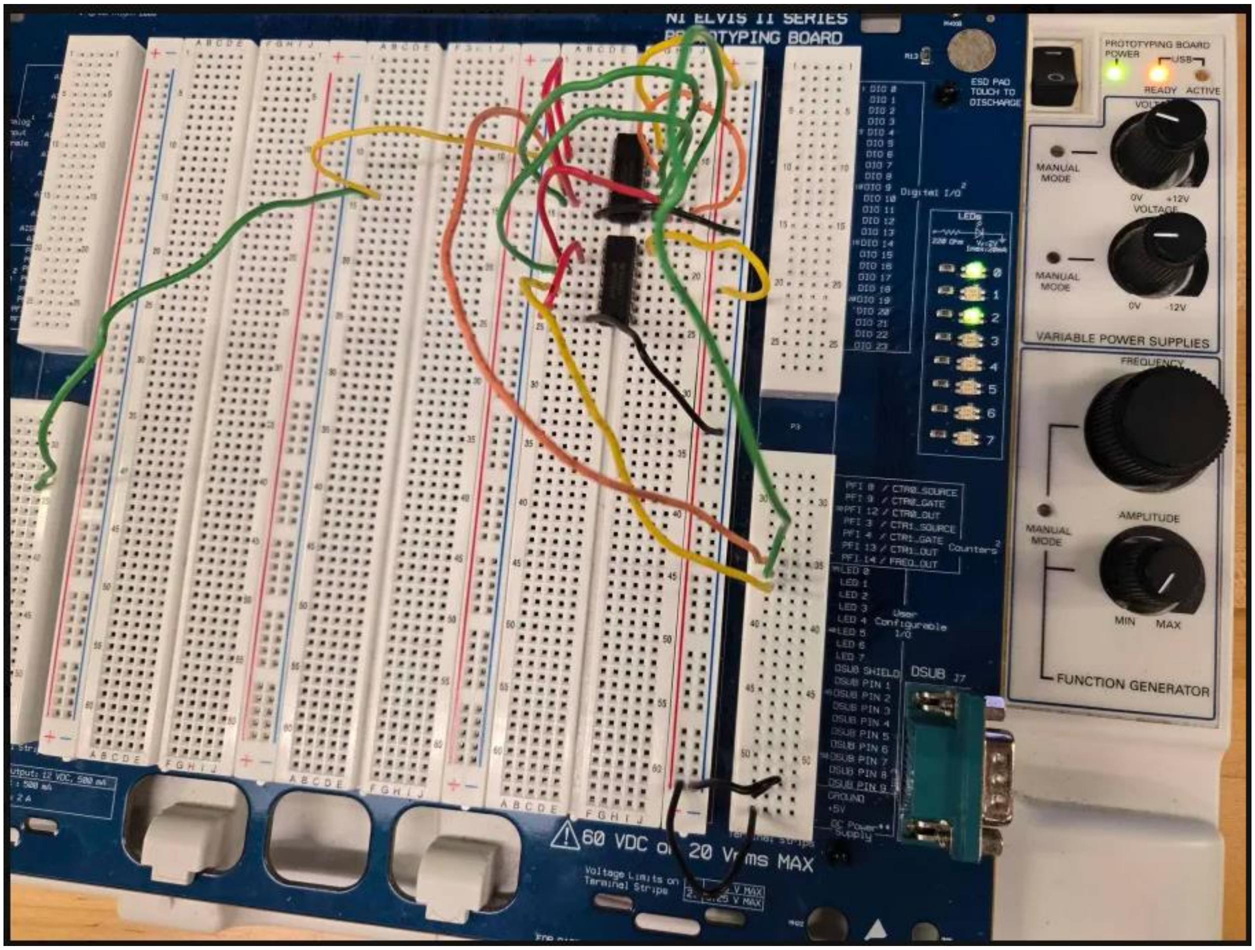

figure 2.6: asynchronous circuit showing a 5.

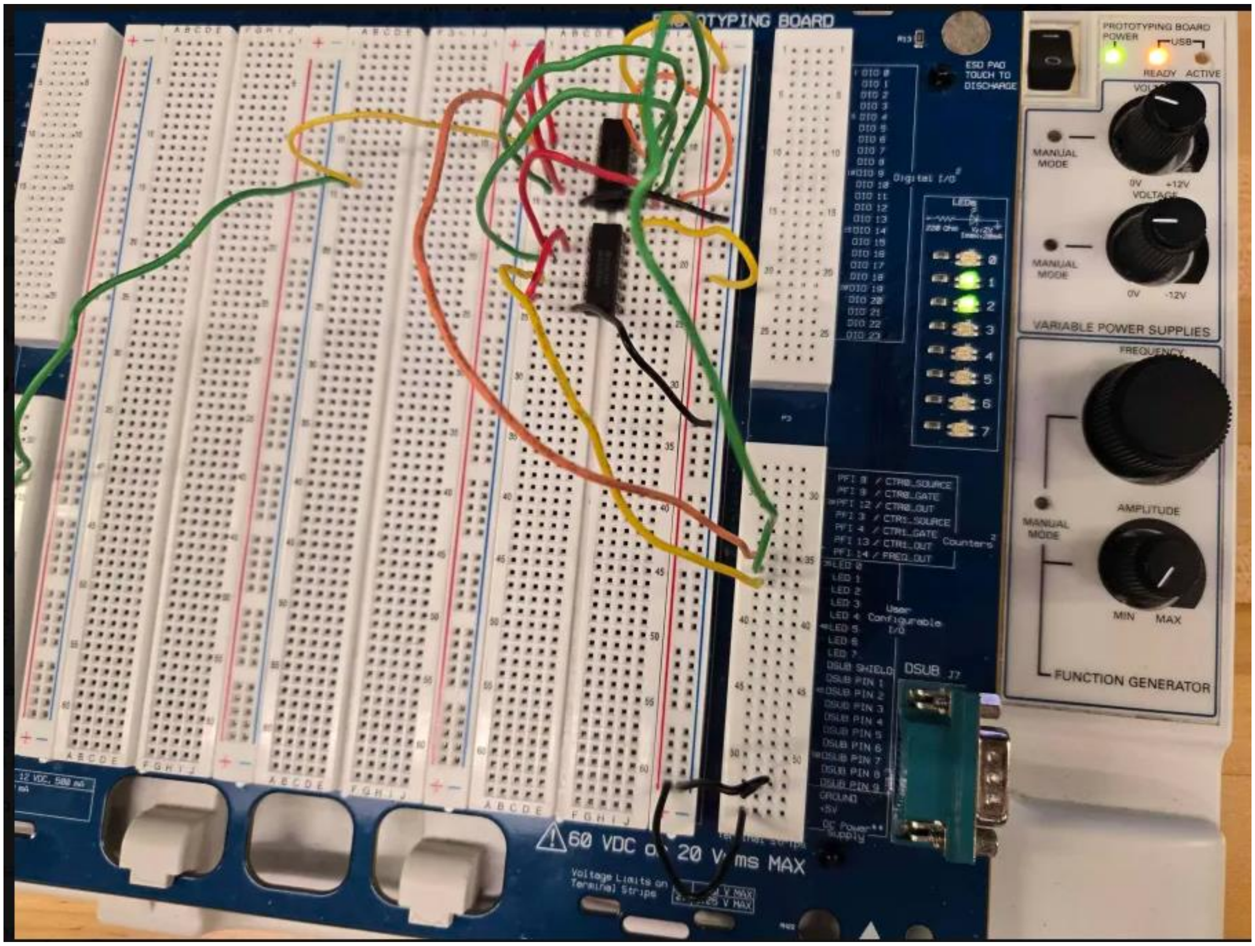

figure 2.7: asynchronous circuit showing a 6.

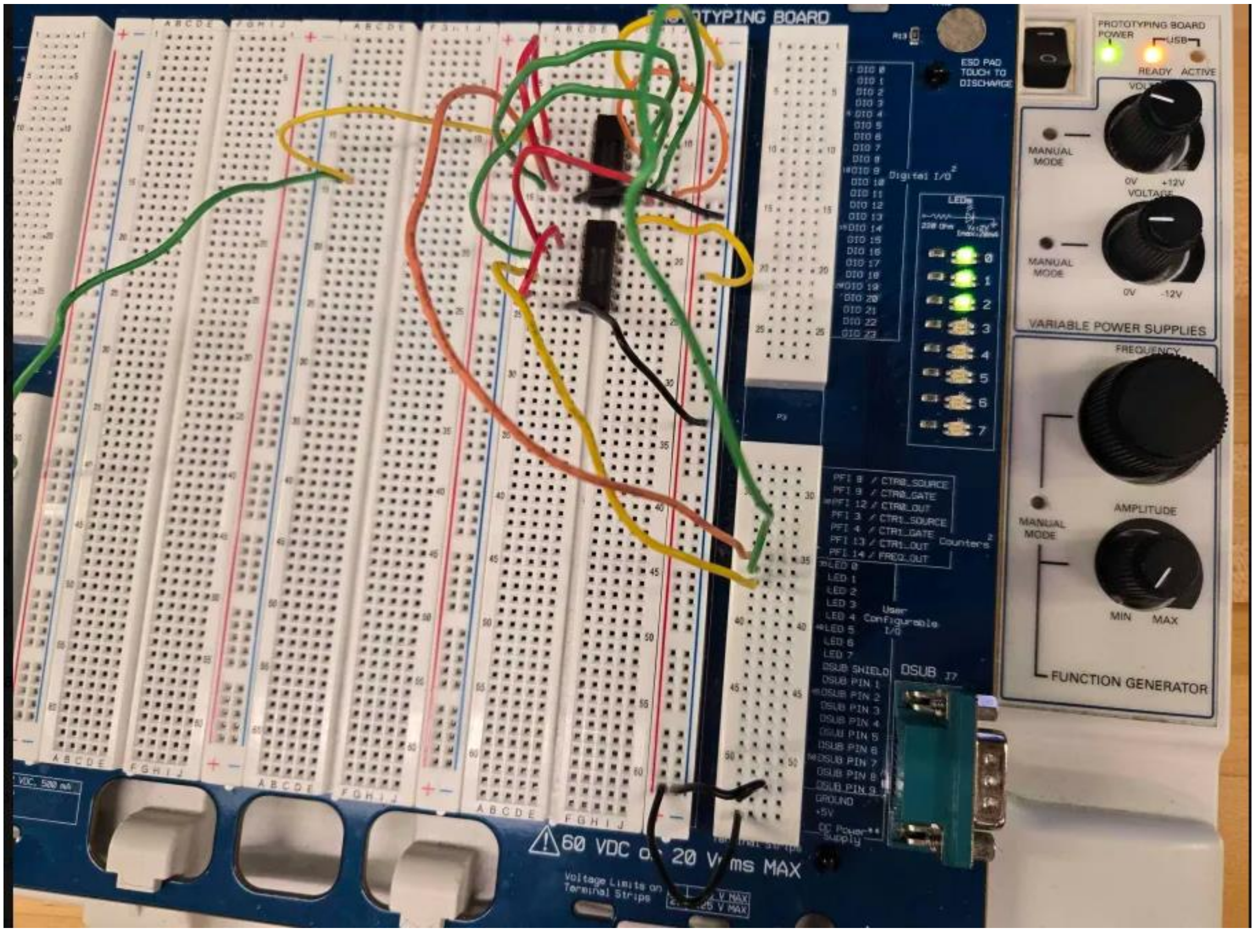

figure 2.8: asynchronous circuit showing a 7.

Conclusion

This project was designed to understand that the how an asynchronous and synchronous circuit can work to make a counter using D-flipflop. This project also helped understand how these circuits are wired and to understand when the clock signal is needed to be given to each D-flip-flop vs only when given to the fist flip-flop.